SmartCity AR

Re-imagining Cape Town's Foreshore Freeway

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2025 Honours Project

Table of Contents

Project Introduction

Problem
- Hard to visualize planning proposals
- 2D plans lack context and interactivity

Solution
- AR app for real-world infrastructure visualization

Tools
- Unity
- AR Foundation
- Low-poly models

Goal
- Support informed planning decisions

High-Level Architectural Design

Modular Architecture

•Ensures scalability & reusability

Core Components

•**UI:** User interaction & commands

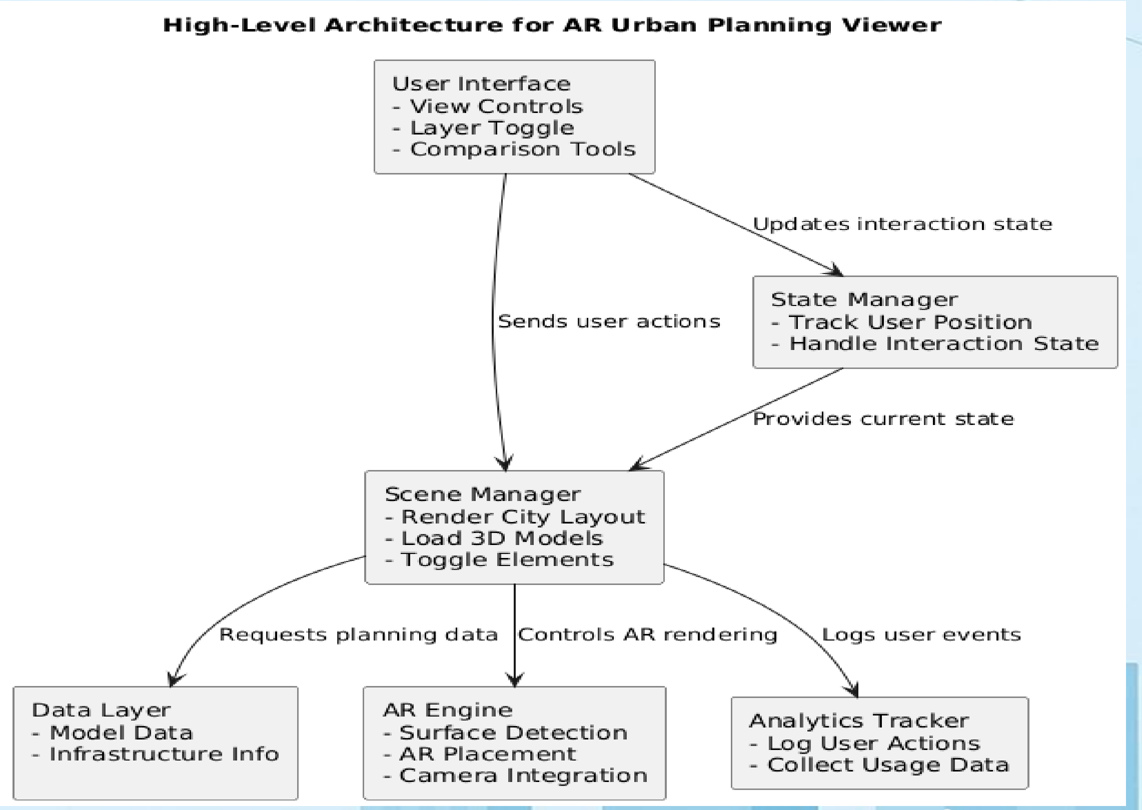
•**Scene Manager:** Renders AR view, controls elements

•**Data Layer:** Holds model & planning data

•**AR Engine:** Handles placement (Unity AR Foundation)

•**State Manager:** Tracks user interaction state

•**Analytics Tracker:** Logs usage behavior



Key Flow

UI → Scene Manager → AR Engine / Analytics

Interface Design

Buttons are placed at the top of the screen for easy access and quick interaction without obstructing the AR view.

Clean & Intuitive Layout

- Toggle infrastructure layers
- Compare sustainable vs. unsustainable views
- Navigate with zoom, pan, and rotate
- Visual feedback on interactions

Tools Used

- Unity 2022.60.f1
- AR Foundation
- UI Toolkit

Reset button to reset the scene

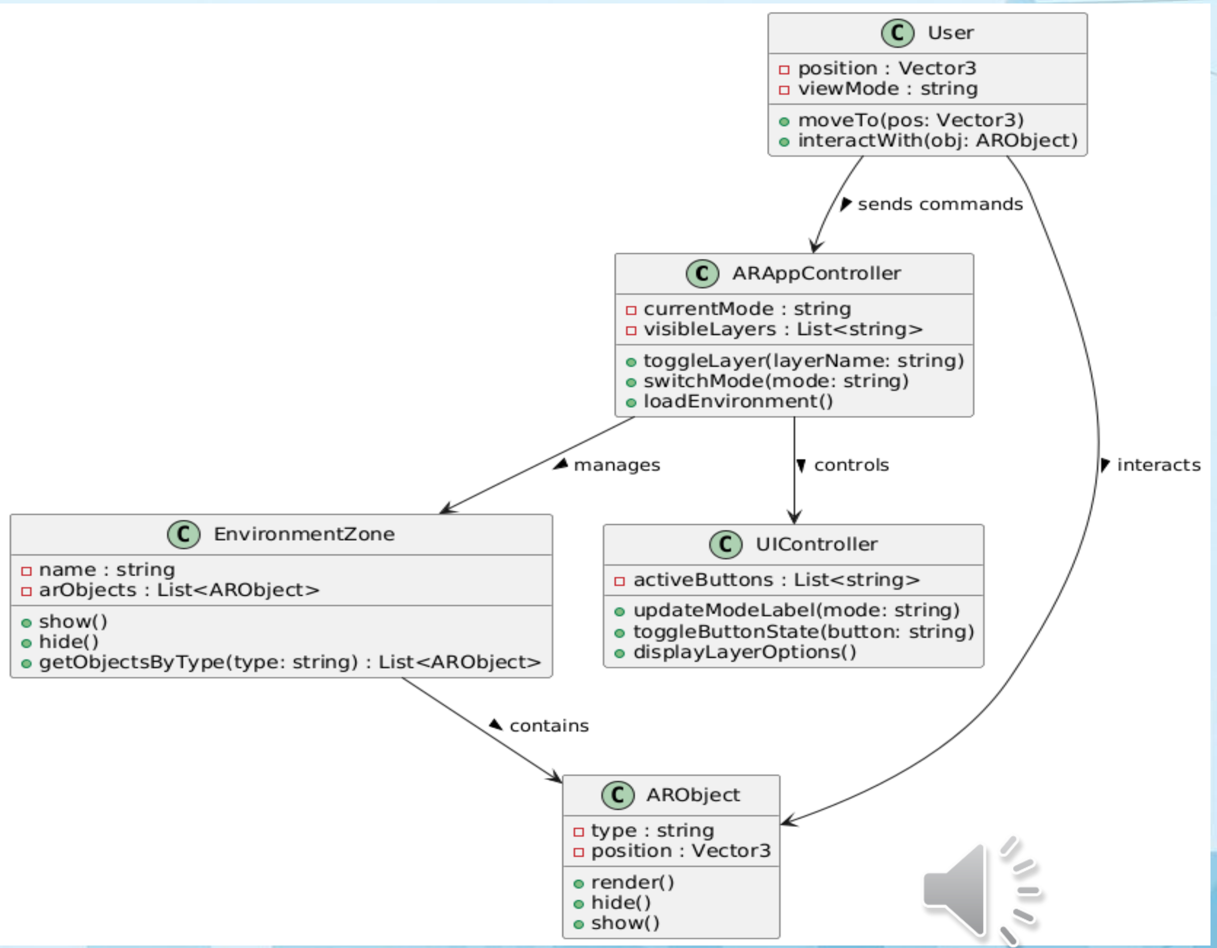
Object-Oriented Design – Class Diagram

<u>Key Classes & Relationships</u>

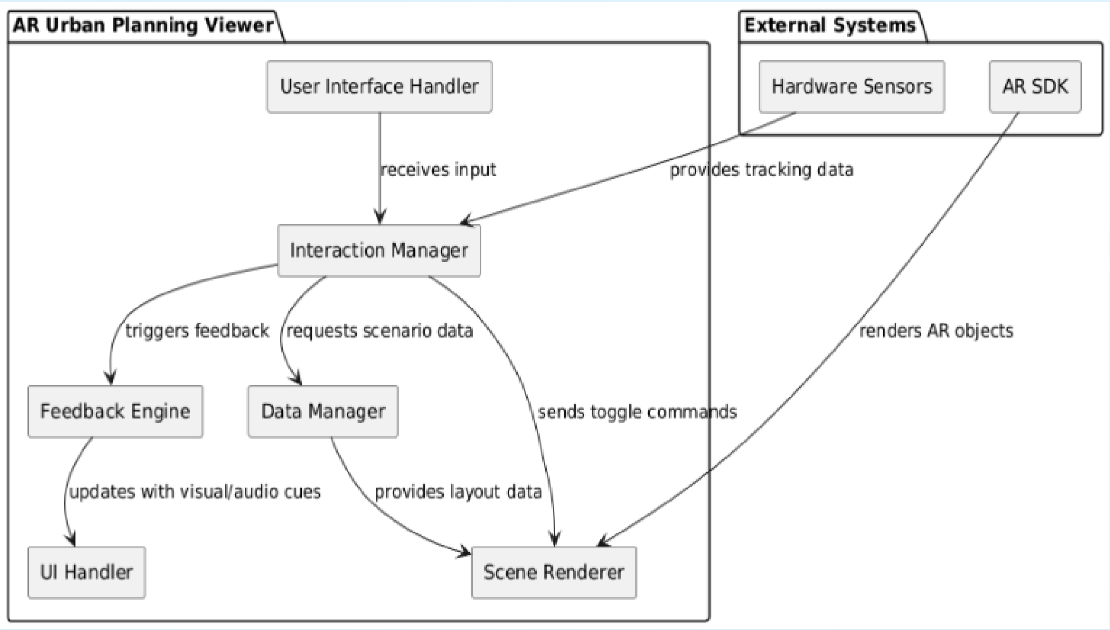
- **ARAppController** – Manages user commands and layer visibility
- **EnvironmentZone** – Groups AR objects by zone
- **ARObject** – Generic AR element with position/rendering logic
- **User** – Simulates user actions (e.g., interaction, movement)
- **UIController** – Handles buttons and labels

<u>Class Relationships</u>

- ARAppController connects UIController + EnvironmentZone
- User interacts with ARObjects via ARAppController
- EnvironmentZones contain multiple ARObjects



Component Level Design



Main Components

- **UI Handler** – Captures input
- **Interaction Manager** – Processes input, updates scene
- **Scene Renderer** – Displays city layout
- **Feedback Engine** – Provides visual/audio cues
- **Data Manager** – Supplies scenario data
- **AR SDK & Sensors** – Handle tracking & placement

Flow
User → UI Handler → Interaction Manager → Scene & Feedback

Prototype

Key Interactions:

- Toggle visibility of:
  - Water & electricity layers
  - MyCiTi bus lanes
  - Sustainable vs. unsustainable scenarios
- Unity Camera movement via Unity touch gestures (pan, zoom, rotate)

AR Features:

- Interactive urban infrastructure exploration
- Built in Unity 2022.3.60f1 with AR Foundation

Conclusion & Future Work

• Improves urban planning understanding through AR

Future additions:

• Real-time traffic & data overlays

• Multi-user collaboration

• Enhanced stakeholder interaction

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